macOS BYOD Management — Account-Driven User Enrollment, App Protection & Conditional Access

Implementation Solution Guide

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Contents

Executive Summary — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access 5

1. Solution Overview — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access 5

What this solution is . . . 5

The honest macOS app-protection note (read this first) . . . 6

The three BYOD models for Macs (choose deliberately) . . . 6

How account-driven user enrollment works (Model A, step by step) . . . 7

Benefits . . . 8

Limitations (be honest with buyers) . . . 8

The standard deployment model . . . 8

2. Requirements — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access 9

R1. Microsoft Intune with macOS enrollment allowed . . . 9

R2. Account-driven user enrollment supported OS floor . . . 9

R3. Managed Apple Account capability (Apple identity for user enrollment) . . . 9

R4. macOS BYOD compliance policy (scoped to the pilot) . . . 10

R5. Conditional Access capability (Entra ID P1 or higher) . . . 10

R6. Break-glass (emergency access) accounts . . . 10

R7. Pilot user identities and their personal Macs . . . 10

R8. User communication / consent for BYOD . . . 10

R9. Network access to Microsoft and Apple endpoints . . . 11

R10. Administrative roles for enrollment, compliance, and Conditional Access . . . 11

Global Requirements — Intune Solutions (category-level) 11

G1. Microsoft Entra ID tenant . . . 11

G2. Microsoft Intune subscription . . . 11

G3. Administrative roles . . . 11

G4. Device enrollment in Intune . . . 12

G5. Pilot security group . . . 12

G6. Supported Windows baseline . . . 12

G7. Network access to Microsoft cloud endpoints . . . 12

3. Implementation — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access 12

Phase 0 — Assess (confirm readiness) . . . 13

Phase 1 — Configure account-driven user enrollment . . . 13

Phase 2 — Scope BYOD compliance . . . 14

Phase 3 — Build the Conditional Access gate . . . 14
Phase 4 — Pilot . . . 15
Phase 5 — Validate . . . 15
Phase 6 — Expand . . . 16
5. Testing & Validation — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access 16
Test 1 — Tenant readiness . . . 16
Test 2 — Personal Mac enrolls (account-driven user enrollment) . . . 17
Test 3 — Managed scope is isolated (privacy) . . . 17
Test 4 — Compliance signal reports . . . 17
Test 5 — Conditional Access (report-only) predicts the right outcome . . . 17
Test 6 — Enforce the gate for the pilot . . . 17
Test 7 — App-protection expectation is correct (honesty check) . . . 17
Test 8 — Selective removal on off-boarding . . . 18
Acceptance gate . . . 18
6. Troubleshooting — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access 18
The "Sign in to Work or School account" option doesn't enrol / no managed scope appears . . . 18
The whole Mac got managed / personal data looks managed . . . 18
Device enrolled but **no compliance state** / shows Not evaluated . . . 19
Device shows **Noncompliant** . . . 19
Conditional Access blocks a legitimate compliant Mac . . . 19
Conditional Access lets an **unmanaged** Mac through . . . 19
Locked out after enabling the CA policy . . . 19
"But I expected App Protection Policies to cover the Mac apps" . . . 19
Off-boarding wiped the whole Mac / affected personal data . . . 20
Useful checkpoints . . . 20
7. Best Practices — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access 20
Choose the least footprint that still protects data . . . 20
Be honest about macOS app protection (the #1 misconception) . . . 21
Privacy: earn and keep user trust . . . 21
Conditional Access discipline (don't lock people out) . . . 21
Operating system & compliance . . . 22
Operations & documentation . . . 22
Security posture . . . 22
8. Deliverables Summary — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access 22

Documentation (this guide) 22
PowerShell scripts (Microsoft Graph) — `MACBYOD-Scripts.zip` 23
Named objects this solution creates 24
Hero image 24
How it's delivered 24
Appendix A — PowerShell Scripts 25

Executive Summary — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access

Your people use their **own** Macs to reach company email, files, and apps. You need that data protected — but you don't own the device, and heavy-handed management will (rightly) be rejected. This solution builds a **macOS BYOD strategy** in **Microsoft Intune** and **Microsoft Entra ID** that gets the balance right: **the lightest management footprint that still protects data, and an access gate that turns away anything unmanaged or unhealthy.**

It is written so a single administrator can follow it end to end:

- **Right-sized enrollment.** We standardize on Apple's **account-driven user enrollment** — the user

signs in with their Entra ID and macOS creates a **managed Apple Account** and an **isolated managed volume**. Corporate apps and data are protected there; the user's personal apps, data, and Apple ID are never touched. We also document the two alternatives (full **device enrollment**, and **unmanaged + Conditional Access**) so you choose the model deliberately.

- **Access gated on device health.** A named **Conditional Access** policy —

`CA - Require Compliant Mac - Pilot` — allows Microsoft 365 access only from a **compliant, enrolled** Mac, and blocks the rest. Deployed **report-only first**, with break-glass accounts excluded, so no one gets locked out.

- **An honest app-protection model.** The most expensive mistake in macOS BYOD is assuming Intune **App

Protection Policies (MAM) **work like they do on iOS/Android.** They do not **— there is no iOS-style per-app container for arbitrary Mac apps. On macOS, corporate-data protection comes from** MDM scope + compliance + Conditional Access**. We state this plainly and build the protection the way macOS actually supports it.

The package follows the standard repeatable shape — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — with exact, click-by-click Intune and Entra steps, a requirements guide that tells you how to verify *and* fix each prerequisite, five ready-to-run PowerShell scripts (Microsoft Graph; discovery/validation/ report are read-only, remediation/scope-targeting default to `-WhatIf`), and testing/troubleshooting guides with clear pass/fail gates.

Outcome: company data protected on personal Macs — the right enrollment model and a Conditional Access gate, without over-managing the user's device — proven on a pilot before you scale, delivered by your own team without a consulting engagement.

1. Solution Overview — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access

What this solution is

Your people want to reach company email, files, and apps from their **own** Macs. You need that data protected without seizing control of a device you don't own. This solution builds a **macOS BYOD (Bring-Your-Own-Device) strategy** in **Microsoft Intune** and **Microsoft Entra ID** that does three things, in the right order:

1. **Enroll personal Macs with the lightest footprint that still gives you control** — Apple's

account-driven user enrollment, which manages a *managed Apple Account* and corporate apps/data on the Mac while leaving the user's personal apps, data, and identity untouched.

1. **Gate access on device state** — a named **Conditional Access** policy that only lets a Mac reach

Microsoft 365 when it is **compliant** (and enrolled), so an unmanaged or unhealthy Mac is turned away.

1. **Tell the honest truth about "app protection" on macOS** — because the biggest mistake buyers make

here is assuming macOS works like iOS/Android. It does not (see the honest note below).

The result: company data is protected on non-corporate Macs, using the right enrollment model and a compliance gate — deployed pilot-first and rollback-ready — **without over-managing the user's device**.

The honest macOS app-protection note (read this first)

On **iOS and Android**, Intune **App Protection Policies (APP / "MAM")** can protect corporate data *inside* individual apps (Outlook, Teams, Edge) — encryption, copy/paste limits, a PIN, and selective wipe of company data — **without enrolling the device at all**. It is tempting to assume the same is available for BYOD Macs. **It is not.**

- **Intune App Protection Policies do NOT cover macOS apps the way they do iOS/Android.** There is no

equivalent per-app "MAM without enrollment" containerization for arbitrary Mac apps. The APP policy types you see for iOS/Android are not the macOS story.

- On **macOS**, corporate-data protection comes from a **different place**: the device being **enrolled

and in scope for MDM **(so you can push configuration, encryption, and compliance signals)** plus Conditional Access** (so only a compliant/enrolled Mac is allowed to reach the data in the first place).

- In practical terms: **the protection boundary on a BYOD Mac is the enrollment + compliance +

Conditional Access combination, not an iOS-style in-app container. **If you need data protected on a Mac, the Mac must be** enrolled **and** compliant**, and access must be** gated**.

Everywhere this guide says "protect company data on a personal Mac," that is what it means. We will not overclaim macOS MAM, and neither should your internal messaging to users.

The three BYOD models for Macs (choose deliberately)

There is no single "BYOD" setting — there are three postures, each with a different management footprint and a different protection story. Pick per your risk appetite; this solution **standardizes on account-driven user enrollment** as the default, with the other two documented so you choose with eyes open.

Model A — Account-driven user enrollment (this solution's default)

• **What it is:** The user opens **Settings** → **(their name area)** → **Sign in to Work or School account** and signs in with their **Entra ID**. macOS creates a **managed Apple Account** and a **separate, cryptographically isolated APFS volume** for managed data; corporate apps and data live there, personal data stays personal.

• **Footprint: Light.** No supervision, no device serial required, no wipe of the whole Mac — only the **managed** data can be removed. Users keep their personal Apple ID and privacy.

• **What you get:** MDM management of the *corporate* scope, a compliance signal, and Entra device registration — enough to drive **Conditional Access**.

• **Best for:** Personal Macs where you want data protected and access gated, but must respect that the device is the user's own.

Model B — Device enrollment (BYOD, but fuller management)

• **What it is:** The user enrolls the whole Mac via the **Company Portal** app (Intune's standard macOS device enrollment). The **entire device** comes under MDM.

• **Footprint: Heavier.** You manage the whole Mac (not just a corporate volume). It is *not* supervised (that requires company ownership / ADE), but you can push broader configuration and, if needed, wipe the device.

• **Best for:** Personal Macs where the user accepts fuller management in exchange for full access — or where user enrollment's constraints don't fit. More footprint than most BYOD users will accept.

Model C — Unmanaged + Conditional Access only (no enrollment)

• **What it is:** The Mac is **not enrolled** at all. You lean on **Conditional Access** (e.g. require MFA, or restrict to browser-only sessions with app-enforced restrictions) to limit what an unmanaged Mac can do.

• **Footprint: None on the device** — but also **no device-health signal**, so you cannot require *compliance* (there's nothing reporting it). Protection is limited to identity-layer controls.

• **Best for:** Low-risk access, contractors, or a stopgap. **Cannot** satisfy a "require compliant device" policy, because an unmanaged Mac has no compliance state.

> ***This solution builds Model A*** *(account-driven user enrollment) and gates it with Conditional Access requiring* ***compliant*** *(and enrolled) Macs. Model C is covered so you understand what you give up by *not* enrolling; Model B is the fallback when user enrollment doesn't fit.*

How account-driven user enrollment works (Model A, step by step)

1. **Tenant readiness.** macOS enrollment is allowed, and account-driven **user** enrollment is enabled as the platform's enrollment type (Implementation §Phase 1). You do **not** need Apple Business Manager for this model — it is identity-driven, not device-serial-driven.

1. **User initiates from the Mac.** In **System Settings**, the user adds their **Work or School

account** (their Entra ID). This is user-initiated by design — you are not seizing the device.

1. **Managed Apple Account + isolated volume.** macOS provisions a **managed Apple Account** and a **separate managed APFS volume**. Corporate apps/data are placed there; personal data is not touched.

1. **MDM applies the corporate scope.** Intune pushes the configuration and compliance policy scoped to

BYOD, and the Mac reports a **compliance state**. The Mac registers with **Entra ID** as a device.

1. **Conditional Access gates access.** Your named CA policy checks: is this Mac **compliant** (and enrolled)? If yes, access to Microsoft 365 is allowed; if no, it is blocked or challenged.

1. **Un-enrollment = selective removal.** When the user removes the work account (or you retire it), only the **managed** volume/data is removed — the personal Mac stays intact.

Benefits

- **Least footprint that still protects data** — manage the corporate scope, not the user's whole Mac.
- **Access gated on device health** — Conditional Access allows only compliant, enrolled Macs.
- **Privacy-respecting** — personal apps, data, and Apple ID stay the user's; only managed data is ever removed.
- **No Apple Business Manager required** for the user-enrollment model — identity-driven onboarding.
- **Honest, defensible model** — you protect data the way macOS actually supports it, not with an iOS-style promise you can't keep.
- **Pilot-first and rollback-ready** — every step is scoped to a pilot group and reversible.

Limitations (be honest with buyers)

- **No iOS-style app protection on macOS.** As above — protection is enrollment + compliance + Conditional Access, **not** per-app MAM containers for arbitrary Mac apps.
- **Account-driven user enrollment requires a recent macOS.** Account-driven **user** enrollment is a modern macOS capability; older Macs may only support device enrollment (Model B). Confirm your OS floor in Requirements (R-OS) and flag anything below it rather than assuming.
- **User-initiated by design.** You cannot silently push user enrollment onto a personal Mac — the user adds the work account. Your rollout must include user communication.
- **Unmanaged Macs can't be "compliant."** If you choose Model C, a require-compliance CA policy will block those Macs — that is expected, not a bug.
- **Conditional Access can lock people out if misconfigured.** Always deploy CA in **report-only** first, with break-glass accounts excluded (Best Practices).

The standard deployment model

This solution uses **one** primary model so the process is repeatable across our catalog:

> ***Personal (BYOD) Macs onboarded with account-driven user enrollment into Microsoft Intune, scoped to a BYOD compliance policy, and gated by a named Conditional Access policy that requires a compliant (enrolled) Mac — piloted, validated, then expanded.***

Every step follows the repeatable pattern: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

2. Requirements — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access

Confirm **every** requirement below before you begin. Each entry follows the same pattern: **Description · Why required · How to verify · How to implement if missing.** Category-wide prerequisites (tenant, Intune licensing, admin roles, pilot group, network) are covered once in the **Global Requirements — Intune Solutions** document; the items here are specific to macOS BYOD account-driven user enrollment and the Conditional Access gate.

All admin-center steps use the Microsoft Intune admin center at `https://intune.microsoft.com` and Microsoft Entra at `https://entra.microsoft.com` unless noted.

R1. Microsoft Intune with macOS enrollment allowed

Description: An Intune-licensed tenant where the **macOS** platform is permitted to enroll. **Why:** An enrollment restriction that blocks macOS will silently stop BYOD Macs from onboarding. **Verify:** Intune → **Devices** → **Enrollment** → **Enrollment device platform restrictions** → **macOS** shows **Personally owned devices = Allow**. **Implement:** Edit the default (or a pilot-targeted) restriction to **Allow** macOS personally owned devices for the pilot group.

R2. Account-driven user enrollment supported OS floor

Description: Target Macs run a macOS version that supports **account-driven user enrollment**. **Why:** Account-driven **user** enrollment is a modern macOS capability; older Macs may only support device enrollment (Model B). Do not assume — confirm and flag anything below the floor. **Verify:** On the Mac, **Apple menu** → **About This Mac**; or after onboarding, Intune → device → **Hardware**. *Uncertain about the exact minimum for your fleet? Flag it and confirm against Microsoft's current macOS enrollment documentation rather than guessing — the floor has moved across macOS releases.* **Implement:** Update the Mac via **System Settings** → **General** → **Software Update**, or set a minimum-OS **compliance policy** so Macs below the floor are flagged. Where a Mac cannot meet the floor, fall back to **device enrollment** (Overview, Model B).

R3. Managed Apple Account capability (Apple identity for user enrollment)

Description: The tenant/domain can issue or federate a **managed Apple Account** used by account-driven user enrollment for the corporate scope. **Why:** Account-driven user enrollment provisions a **managed Apple Account** and an isolated managed volume; the user's **personal** Apple ID is left alone. Without a managed Apple Account path, the corporate scope cannot be established. **Verify:** Confirm your

organization's managed Apple Account approach (federated with Entra, or issued) is in place for the pilot users. *If you are unsure whether managed Apple Accounts are configured for your org, flag it and confirm before piloting — do not invent a state that isn't there.* **Implement:** Establish managed Apple Accounts per Apple's current guidance (federation with Microsoft Entra ID is the common path). Coordinate with whoever owns your Apple identity relationship.

R4. macOS BYOD compliance policy (scoped to the pilot)

Description: An Intune **compliance policy** for macOS defining what "healthy enough for access" means for BYOD (e.g. minimum OS, disk encryption where applicable, no jailbreak/gatekeeper issues). **Why:** Conditional Access can only require a **compliant** Mac if a compliance policy exists and is **assigned** to the Mac. No policy → no compliance signal → the CA gate cannot use it. **Verify:** Intune → **Devices** → **Compliance policies**; a macOS policy named `macOS BYOD Compliance - Pilot` exists and is assigned to your pilot group. **Implement:** Create it during deployment (Implementation §Phase 2), or start from the settings there and assign it to `BYOD-Mac-Pilot`.

R5. Conditional Access capability (Entra ID P1 or higher)

Description: A licensing plan that includes **Conditional Access** (Entra ID P1, included in Microsoft 365 E3/E5 and Business Premium). **Why:** The access gate — "require a compliant/enrolled Mac" — is a Conditional Access policy. **Verify:** Entra → **Protection** → **Conditional Access** loads and lets you create policies; **Entra** → **Overview** → **Licenses** shows a P1 (or higher) plan. **Implement:** Assign an Entra ID P1-bearing license (E3/E5/Business Premium) to the admins and the pilot users in scope.

R6. Break-glass (emergency access) accounts

Description: Two cloud-only **emergency access** accounts excluded from Conditional Access. **Why:** A misconfigured CA policy can lock **everyone** out. Break-glass accounts are your way back in. **Verify:** Entra → **Users**; two dedicated emergency accounts exist and are **excluded** from all CA policies (Entra → Protection → Conditional Access → each policy → Users → Exclude). **Implement:** Create two cloud-only accounts (long random passwords, stored securely, MFA per Microsoft guidance), and **exclude** them from every CA policy before you enable any policy.

R7. Pilot user identities and their personal Macs

Description: A small set of **pilot users** who will onboard their **own** Macs, and a Microsoft Entra security group named `BYOD-Mac-Pilot` containing them. **Why:** Account-driven user enrollment is **user-initiated**; you need willing pilot users who will add their work account on a real personal Mac to validate the end-to-end experience. **Verify:** Entra → **Groups** → `BYOD-Mac-Pilot` exists with the pilot users as members. **Implement:** Entra → **Groups** → **New group** → **Security**, name it `BYOD-Mac-Pilot`, add the pilot users. (This is the category Global Requirement G5, named for this solution.)

R8. User communication / consent for BYOD

Description: A short user-facing message explaining what account-driven user enrollment does and does **not** do to a personal Mac. **Why:** BYOD only works if users trust it. They must understand the corporate scope is isolated and only managed data is ever removed — and that you cannot see or wipe their personal data. **Verify:** A one-page BYOD explainer exists for pilot users (privacy scope, what's managed,

how to remove the work account). **Implement:** Draft it from the Overview's "How it works" and "privacy" points; the Best Practices guide lists exactly what to tell users.

R9. Network access to Microsoft and Apple endpoints

Description: Outbound HTTPS from the pilot Macs to Microsoft Entra/Intune and to Apple's identity and enrollment endpoints. **Why:** Sign-in, enrollment, managed Apple Account provisioning, policy sync, and Conditional Access evaluation all depend on these endpoints. **Verify:** From the Mac's network, reach `https://login.microsoftonline.com`, `https://enrollment.manage.microsoft.com`, and Apple's identity/enrollment endpoints (e.g. `https://gdmf.apple.com`, `https://identity.apple.com`). **Implement:** Allow Microsoft's published Microsoft 365/Intune/Entra URL & IP ranges and Apple's published enrollment/identity ranges through proxy/firewall. Do **not** TLS-inspect Apple enrollment and push traffic.

R10. Administrative roles for enrollment, compliance, and Conditional Access

Description: The operator holds the roles needed to configure enrollment/compliance (**Intune Administrator**) and Conditional Access (**Conditional Access Administrator** or **Security Administrator**). **Why:** Creating the enrollment type, the compliance policy, and the CA policy requires elevated rights across both Intune and Entra. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator** and a CA-capable role are assigned to the operator. **Implement:** Assign **Intune Administrator** plus **Conditional Access Administrator** (or **Security Administrator**); reserve Global Administrator for tenant-wide changes.

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access

Follow these steps **in order**. The *why* for each capability lives in the Solution Overview; the steps below are deliberately short and exact so you can **click to implement**. Throughout, the repeatable pattern applies: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.** Do everything against the pilot group `BYOD-Mac-Pilot` first.

All admin-center steps use the Microsoft Intune admin center at `https://intune.microsoft.com` and Microsoft Entra at `https://entra.microsoft.com` unless noted. Named objects you will create:

Object	Name	Where
Pilot security group	`BYOD-Mac-Pilot`	Entra → Groups

Object	Name	Where
Enrollment type	**account-driven user enrollment** (macOS)	Intune → Enrollment
Compliance policy	`macOS BYOD Compliance - Pilot`	Intune → Compliance policies
Conditional Access policy	`CA - Require Compliant Mac - Pilot`	Entra → Conditional Access

Phase 0 — Assess (confirm readiness)

Step 1. Run the discovery script

1. On an admin workstation with the **Microsoft Graph PowerShell SDK**, run `MACBYOD-Discovery.ps1` (read-only).
1. Confirm: macOS enrollment **allowed**, an account-driven user-enrollment configuration present (or flagged as not yet enabled), a macOS **compliance policy**, and whether a **Conditional Access** policy already gates macOS. Resolve any FAIL before continuing.

Step 2. Confirm the pilot group exists

1. Entra → **Groups**; confirm `BYOD-Mac-Pilot` exists with your pilot users (Requirements R7).
2. If missing: **Groups** → **New group** → **Security**, name it `BYOD-Mac-Pilot`, add pilot users.

Phase 1 — Configure account-driven user enrollment

Step 3. Allow macOS personally owned enrollment

1. Intune → **Devices** → **Enrollment** → **Enrollment device platform restrictions**.
2. Open the default (or a pilot-targeted) restriction → **macOS**.
3. Set **Personally owned devices = Allow**. Save.

Step 4. Enable account-driven user enrollment for macOS

1. Intune → **Devices** → **Enrollment** → **Apple** → open the **macOS** enrollment area.
2. Select the **account-driven user enrollment** enrollment type (the identity-driven, lighter-footprint BYOD option — **not** device enrollment and **not** Automated Device Enrollment).
1. Configure it to target `BYOD-Mac-Pilot` so only pilot users get this experience first.
2. Save.

> ***If your console does not surface account-driven *user* enrollment for macOS**, confirm your OS floor (Requirements R2) and your managed Apple Account path (R3) — both are prerequisites. **Flag it and confirm against Microsoft's current macOS enrollment documentation rather than forcing a different enrollment type.** Do not silently substitute device enrollment; that changes the footprint the*

buyer asked for.

Step 5. Confirm the managed Apple Account path

1. Verify pilot users can obtain a **managed Apple Account** for the corporate scope (Requirements R3) —

the user's **personal** Apple ID stays untouched.

1. If unclear, coordinate with whoever owns your Apple identity relationship **before** piloting.

Phase 2 — Scope BYOD compliance

Step 6. Create the BYOD compliance policy

1. Intune → **Devices** → **Compliance policies** → **Create policy**.
2. **Platform = macOS**; **Create**.
3. **Name:** `macOS BYOD Compliance - Pilot`.
4. **Compliance settings** — keep BYOD-appropriate and minimal (do not over-manage a personal Mac):

• **Minimum OS version** = your confirmed account-driven-enrollment floor (Requirements R2).

• **System Integrity Protection** required (where supported).

• Disk-encryption / firewall settings **only** where they make sense for BYOD and match your policy.

1. **Actions for noncompliance** — start gentle: **Mark noncompliant** immediately; optionally email

the user. (Avoid retire/wipe actions on personal Macs.)

1. **Assignments** → include `BYOD-Mac-Pilot`.
2. **Review + Create.**

*Prefer automation for scoping? Run `MACBYOD-Set-ScopeTargeting.ps1` to assign the compliance policy (and, later, the CA policy) to a target group. It **defaults to** `-WhatIf` and only writes when you pass `-Apply`.*

Phase 3 — Build the Conditional Access gate

Step 7. Confirm break-glass exclusions exist

1. Entra → **Users**; confirm your two **break-glass** emergency accounts exist (Requirements R6).
2. You will exclude them from the CA policy in the next step. **Never skip this.**

Step 8. Create the named Conditional Access policy (report-only first)

1. Entra → **Protection** → **Conditional Access** → **Create new policy**.
2. **Name:** `CA - Require Compliant Mac - Pilot`.
3. **Users:**

- **Include** → `BYOD-Mac-Pilot`.
- **Exclude** → your two **break-glass** accounts.

1. **Target resources: Cloud apps** → **Include** → **Office 365** (or the specific apps you are gating).
2. **Conditions** → **Device platforms: Include** → **macOS** (so this policy targets Macs only).
3. **Grant: Grant access** → **Require device to be marked as compliant** (this is the compliance gate).

Optionally also require **Microsoft Entra hybrid joined / device to be enrolled** per your model; for BYOD user enrollment, **require compliant** is the primary control.

1. **Enable policy = Report-only.**
2. **Create.**

> *Automating the CA build?* `MACBYOD-Remediation.ps1` *can create/repair this policy and* ***always re-asserts break-glass exclusions****; it* ***defaults to*** `-WhatIf` *and only writes with* `-Apply`*.*

Phase 4 — Pilot

Step 9. Onboard a pilot Mac (user-initiated)

1. As a **pilot user on their own Mac**, open **System Settings**.

2. Add the **Work or School account** (the account-driven user-enrollment sign-in), authenticating with the user's **Entra ID**.

1. macOS provisions the **managed Apple Account** and the **separate managed volume**; the corporate scope is applied. Personal apps, data, and Apple ID are untouched.

1. Wait for Intune policy to sync; the Mac appears in **Intune** → **Devices** → **macOS** as a **personally owned**, user-enrolled device.

Step 10. Confirm the compliance signal

1. Intune → **Devices** → **macOS** → **(the pilot device)** → **Compliance**; confirm it evaluates against `macOS BYOD Compliance - Pilot` and reports **Compliant**.

1. Or run `MACBYOD-Validation.ps1 -Serial the-device-serial` for a per-device PASS/FAIL.

Phase 5 — Validate

Step 11. Validate the end-to-end posture

1. Run `MACBYOD-Validation.ps1` against the pilot Mac: enrolled (**account-driven user enrollment**),** personally owned**,** compliant**, and Entra-registered.

1. Review `CA - Require Compliant Mac - Pilot` in **report-only**: Entra → **Sign-in logs** → filter

to the pilot user/macOS → confirm the policy would have **allowed** the compliant Mac and **blocked** a non-compliant/unenrolled one (test from an unmanaged Mac to see the block *predicted*).

1. Run `MACBYOD-Report.ps1` for the BYOD access-posture snapshot (enrolled vs unmanaged, compliant

vs not, which Macs the CA policy would gate).

Step 12. Turn the gate on for the pilot

1. Once report-only shows the intended outcome with **no surprise blocks** for legitimate users, edit

`CA - Require Compliant Mac - Pilot` → **Enable policy = On**.

1. Re-test with a pilot user: a **compliant** enrolled Mac gets in; a **non-compliant/unenrolled** Mac

is **blocked**. Break-glass accounts remain excluded.

Phase 6 — Expand

Step 13. Widen scope in waves

1. Create a production group (e.g. `BYOD-Mac-Users`) and add the next wave of users.
2. Use `MACBYOD-Set-ScopeTargeting.ps1` (preview with `-WhatIf`, then `-Apply`) to assign the

compliance policy and the **CA policy** to the production group — or clone the pilot CA policy to a production-named policy (e.g. `CA - Require Compliant Mac - Production`) and expand its scope.

1. Keep break-glass exclusions on **every** CA policy.
2. Validate each wave with `MACBYOD-Validation.ps1` / `MACBYOD-Report.ps1` before widening

further. Roll out in waves, never all at once.

5. Testing & Validation — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access

Validate against a **pilot Mac** and the `BYOD-Mac-Pilot` group before expanding. Each test has an explicit **pass/fail gate**. Use the included PowerShell scripts (Microsoft Graph) for tenant-side checks and the on-device steps for the user experience. Keep the Conditional Access policy in **report-only** until Tests 1–6 pass.

Test 1 — Tenant readiness

Procedure: Run `MACBYOD-Discovery.ps1`. **Pass:** macOS enrollment **allowed**; account-driven user enrollment present/enabled (or clearly flagged as not yet enabled); a macOS **compliance policy** (`macOS BYOD Compliance - Pilot`) exists; and the CA policy state is reported. No blocking FAIL. **Fail:** Any prerequisite missing → resolve using the Requirements guide before onboarding devices.

Test 2 — Personal Mac enrolls (account-driven user enrollment)

Procedure: As a pilot user on their **own** Mac, add the **Work or School account** in **System Settings** and complete sign-in. **Pass:** The Mac appears in **Intune** → **Devices** → **macOS** as **personally owned**, enrollment type **account-driven user enrollment**; a **managed Apple Account** and separate managed volume were provisioned; the user's personal Apple ID/data are untouched. **Fail:** No enrollment → confirm macOS is **Allowed** (R1), the OS floor is met (R2), and the managed Apple Account path exists (R3). Do **not** substitute device enrollment to "make it work" — flag it.

Test 3 — Managed scope is isolated (privacy)

Procedure: On the enrolled Mac, confirm corporate apps/data live in the managed scope while personal apps/data remain separate. **Pass:** Managed (corporate) data is present in the managed volume; personal data/apps are unaffected; removing the work account would remove **only** managed data. **Fail:** Personal data appears managed, or the whole device is under management → you likely used **device enrollment** (Model B), not **account-driven user enrollment** (Model A). Re-check Step 4 of Implementation.

Test 4 — Compliance signal reports

Procedure: Intune → **Devices** → **macOS** → **(device)** → **Compliance**, or run `MACBYOD-Validation.ps1 -Serial the-device-serial`. **Pass:** The Mac evaluates against `macOS BYOD Compliance - Pilot` and reports **Compliant**. **Fail:** No compliance state → the policy isn't assigned to `BYOD-Mac-Pilot` (R4); **Noncompliant** → address the failing setting (usually OS version) and re-sync.

Test 5 — Conditional Access (report-only) predicts the right outcome

Procedure: With `CA - Require Compliant Mac - Pilot` in **report-only**, have the pilot user sign in to a gated app (e.g. `https://outlook.office.com`) from the enrolled Mac; then attempt from an **unmanaged** Mac. Review Entra → **Sign-in logs** (or run `MACBYOD-Report.ps1`). **Pass:** The log shows the policy **would allow** the **compliant, enrolled** Mac and **would block** the **unmanaged/non-compliant** Mac. Break-glass accounts are excluded. **Fail:** Would block a legitimate compliant Mac → check the grant control and platform condition (Implementation Step 8); would allow an unmanaged Mac → confirm the grant is **Require compliant** and the macOS platform condition is set.

Test 6 — Enforce the gate for the pilot

Procedure: Set `CA - Require Compliant Mac - Pilot` to **On**. Re-test from a compliant enrolled Mac and from an unmanaged Mac. **Pass:** Compliant enrolled Mac → **access granted**; unmanaged/non-compliant Mac → **access blocked**; break-glass account (excluded) → still able to sign in. **Fail:** Legitimate user locked out → revert to report-only, correct scope/exclusions, and re-test. Use a **break-glass** account if you lose admin access.

Test 7 — App-protection expectation is correct (honesty check)

Procedure: Confirm your internal messaging and controls match how macOS actually protects data. **Pass:** Protection for BYOD Macs is delivered by **enrollment + compliance + Conditional Access** — not by an iOS-style App Protection Policy. No one is relying on macOS "MAM" that doesn't exist. **Fail:** Someone expects per-app containerization/selective in-app wipe on arbitrary Mac apps → correct the expectation (Solution Overview, honest note). Gate access via CA instead.

Test 8 — Selective removal on off-boarding

Procedure: Have a pilot user remove the **Work or School account** (or retire the device in Intune). **Pass:** Only the **managed** volume/data is removed; the personal Mac remains intact and usable; the device drops out of the compliant set and the CA gate would block corporate access. **Fail:** Whole device wiped or personal data affected → the wrong enrollment model was used; re-check that this is **account-driven user enrollment** (Model A).

Acceptance gate

Proceed to **Expand** only when Tests 1–8 pass on at least one pilot Mac, the CA policy enforces the intended allow/block outcome with break-glass excluded, and a real pilot user confirms the privacy boundary (personal data untouched, only managed data removable).

6. Troubleshooting — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access

Work top-down: confirm the **enrollment** (personal Mac onboards), then the **compliance signal**, then the **Conditional Access gate**, and only then the **user experience**. Most failures are macOS blocked by an enrollment restriction, an OS below the account-driven-enrollment floor, a compliance policy not assigned, or a CA policy scoped/graded incorrectly. When in doubt, re-run `MACBYOD-Discovery.ps1` and `MACBYOD-Validation.ps1` after each change.

The "Sign in to Work or School account" option doesn't enrol / no managed scope appears

- **macOS not allowed.** Intune → **Devices → Enrollment → Enrollment device platform restrictions →

macOS**; set** Personally owned devices = Allow** (R1).

- **OS below the floor.** Account-driven **user** enrollment needs a modern macOS; update the Mac (R2).

If unsure of the exact minimum, **flag it and confirm** against Microsoft's current documentation — don't guess.

- **Account-driven user enrollment not enabled.** Confirm the enrollment type is configured for macOS and

targets `BYOD-Mac-Pilot` (Implementation Step 4).

- **Managed Apple Account path missing.** The corporate scope needs a **managed Apple Account** (R3);

coordinate with your Apple identity owner. Do **not** silently switch to device enrollment to force it.

The whole Mac got managed / personal data looks managed

- **Wrong model.** This is the symptom of **device enrollment (Model B)**, not **account-driven user

enrollment (Model A)**. Account-driven user enrollment manages only a** separate managed volume**. Remove the enrollment, correct** Implementation Step 4 **to the account-driven** user** enrollment type, and re-onboard.

Device enrolled but no compliance state / shows Not evaluated

- **Compliance policy not assigned.** Confirm `macOS BYOD Compliance - Pilot` is assigned to `BYOD-Mac-Pilot` (R4). Without assignment there is no signal, and Conditional Access can't require it.
- **Sync pending.** Give Intune time to evaluate, or force a sync; re-run `MACBYOD-Validation.ps1 -Serial the-device-serial`.

Device shows Noncompliant

- **OS below the compliance floor** — update macOS (R2), or lower the floor if BYOD-appropriate.
- **A health setting is failing** (e.g. SIP/encryption) — review the failing setting in Intune → device → **Compliance** and address it; keep BYOD compliance **minimal** so you're not failing personal Macs on corporate-only controls.

Conditional Access blocks a legitimate compliant Mac

- **Grant control mismatch.** The policy must grant on **Require device to be marked as compliant** (Implementation Step 8). If it also requires hybrid-join, a BYOD user-enrolled Mac will fail — hybrid join is not the BYOD control.
- **Platform condition too broad/narrow.** Ensure the **Device platforms** condition includes **macOS** and isn't accidentally excluding it.
- **Not actually compliant yet.** Confirm the Mac reports **Compliant** first (see above); the CA gate is only as good as the compliance signal beneath it.

Conditional Access lets an unmanaged Mac through

- **Grant not enforcing compliance.** Confirm the grant is **Require compliant** and the policy is **On** (not report-only). Remember: an **unmanaged** Mac has **no compliance state**, so a require-compliance policy should block it — if it's getting through, the policy isn't applying (check user/app/platform scope).

Locked out after enabling the CA policy

- **Use a break-glass account.** Sign in with one of your two excluded **emergency access** accounts (R6), set the policy back to **report-only**, fix the scope/exclusions, and re-test. This is exactly why break-glass accounts and report-only-first are mandatory.

"But I expected App Protection Policies to cover the Mac apps"

• **They don't — this is expected, not a bug.** Intune **App Protection Policies (MAM)** cover iOS/Android

apps, **not** arbitrary macOS apps. On macOS the protection boundary is **enrollment + compliance + Conditional Access**. Re-set the expectation (Solution Overview, honest note) and gate access with CA rather than looking for a macOS MAM container that doesn't exist.

Off-boarding wiped the whole Mac / affected personal data

• **Wrong action or wrong model.** For account-driven user enrollment, removing the work account (or retiring the device) removes **only** the managed data. If a full wipe happened, a device-level retire/wipe action was used or the device was **device-enrolled** — avoid wipe actions on BYOD and confirm the enrollment model.

Useful checkpoints

• **Intune:** Devices → macOS → (device) → **Hardware / Compliance / Configuration**; Enrollment → platform restrictions; Compliance policies → assignment.

• **Entra:** Protection → **Conditional Access** → policy state (report-only vs on) and user/exclusion scope; **Sign-in logs** → filter to the pilot user/macOS to see which policies applied and the result.

• **On the Mac: System Settings** → **General** → **Device Management / Profiles** (managed scope present);

the user's **Work or School account** entry.

• **Scripts:** re-run `MACBYOD-Validation.ps1` after each change to confirm the gate; `MACBYOD-Report.ps1`

for the fleet-wide posture.

7. Best Practices — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access

Field-proven guidance to keep the deployment reliable, private, and low-footprint — the two things that make macOS BYOD succeed or fail are **respecting the user's device** and **being honest about what protection macOS actually provides**.

Choose the least footprint that still protects data

• **Default to account-driven user enrollment (Model A).** It manages only the corporate scope, leaves personal data alone, and still yields a compliance signal for Conditional Access. Use the heavier **device enrollment (Model B)** only when user enrollment genuinely doesn't fit.

• **Don't over-manage a personal Mac.** Keep the BYOD compliance policy **minimal** — the OS floor and a

few sensible health checks. Every additional control on someone's own Mac erodes trust and adoption.

- **Never use retire/wipe-the-device actions on BYOD.** For account-driven user enrollment, off-boarding

removes only the **managed** data by design — lean on that, not a full wipe.

Be honest about macOS app protection (the #1 misconception)

- **Do not promise iOS-style App Protection Policies (MAM) on macOS — they don't exist for arbitrary Mac

apps. **Corporate-data protection on a Mac comes from** enrollment + compliance + Conditional Access**, not per-app containers or selective in-app wipe.

- **Message this to stakeholders up front.** If a security requirement assumes macOS MAM, correct it

before you build — otherwise you'll ship a control that can't do what someone expects.

- **Gate with Conditional Access instead.** The reliable macOS control is "only a compliant, enrolled Mac

reaches the data." That *is* the protection boundary.

Privacy: earn and keep user trust

- **Publish a one-page BYOD explainer** (Requirements R8): what's managed (corporate scope only), what

IT can and **cannot** see or do (no personal data, no personal Apple ID, no whole-device wipe), and how the user removes the work account.

- **Make removal easy and honest.** Tell users exactly how to remove the work account and what happens

(only managed data goes). Ease of exit increases willingness to enrol.

- **Keep the managed Apple Account path clean.** The corporate scope uses a **managed Apple Account**;

never ask users to hand over their personal Apple ID.

Conditional Access discipline (don't lock people out)

- **Always start in report-only.** Deploy `CA - Require Compliant Mac - Pilot` in **report-only**, read

the sign-in logs, and only enforce once the predicted allow/block matches intent.

- **Exclude two break-glass accounts from every CA policy — always.** Verify the exclusion before you

enable any policy; a misconfigured require-compliance policy can lock out the whole tenant.

- **Scope by platform.** Set the **macOS** device-platform condition so this policy affects Macs only and

doesn't collide with your Windows/iOS gates.

• **Expand in waves.** Move from `BYOD-Mac-Pilot` to a production group deliberately; re-validate each wave with the scripts before widening.

Operating system & compliance

• **Confirm and pin your OS floor.** Account-driven user enrollment needs a modern macOS; set a minimum-OS compliance setting so Macs below the floor are flagged rather than silently failing to enrol. If you're unsure of the exact minimum, **flag it and confirm** — don't guess.

• **Keep compliance BYOD-appropriate.** Require what protects data (OS floor, SIP, encryption where it makes sense) — not a full corporate lockdown of a personal machine.

Operations & documentation

• **Keep a runbook** of the named objects (`BYOD-Mac-Pilot`, `macOS BYOD Compliance - Pilot`, `CA - Require Compliant Mac - Pilot`), the managed Apple Account owner, the OS floor, and the break-glass account locations.

• **Report monthly** with `MACBYOD-Report.ps1`: enrolled BYOD Macs vs unmanaged, compliant vs not, and

which Macs the CA policy gates. A rising "unmanaged but accessing" count is a signal to act.

• **Use `-WhatIf` first on every change.** `MACBYOD-Remediation.ps1` and `MACBYOD-Set-ScopeTargeting.ps1`

default to preview mode; review the intended change, then re-run with `-Apply`.

Security posture

• **Combine BYOD with identity hardening** (MFA, sign-in risk policies) so the CA gate is one layer of several, not the only one.

• **Treat unmanaged Macs honestly.** If you allow Model C (unmanaged + CA), remember those Macs **cannot

be compliant** — a require-compliance policy will block them. Decide that intentionally.

8. Deliverables Summary — macOS BYOD Management: Account-Driven User Enrollment, App Protection & Conditional Access

Everything included in this solution package, and what each piece is for.

Documentation (this guide)

• **Executive Summary** — the business case and outcome in one page.

• **Solution Overview** — the three BYOD models for Macs (account-driven **user enrollment** vs

device enrollment vs **unmanaged + Conditional Access**), the management footprint each carries, how account-driven user enrollment works step by step, and the **honest note** that Intune App Protection Policies do **not** cover macOS apps the way they do iOS/Android — protection comes via MDM scope + Conditional Access.

• **Requirements Guide** — every prerequisite (Intune with macOS enrollment allowed, the OS floor, the

managed Apple Account path, a BYOD compliance policy, Conditional Access licensing, break-glass accounts, pilot users/group, user communication, network, admin roles) with how to **verify** and how to **implement if missing**, plus the category-level Global Requirements.

• **Implementation Guide** — exact, ordered, click-by-click steps across six phases: assess, enable

account-driven user enrollment, scope BYOD compliance, build the named Conditional Access policy (require compliant/enrolled Mac), pilot, validate, and expand.

• **Testing & Validation** — eight tests with explicit pass/fail gates (personal Mac enrolls, managed

scope isolated, compliance reports, CA predicts then enforces the right allow/block, the honesty check, and selective removal) and an acceptance gate.

• **Troubleshooting** — top-down diagnosis for enrollment, compliance, and Conditional Access failures.

• **Best Practices** — least footprint, privacy/user trust, the macOS app-protection honesty rule, and

Conditional Access discipline.

PowerShell scripts (Microsoft Graph) — `MACBYOD-Scripts.zip`

All scripts are tenant-side (run from an admin workstation with the Microsoft Graph PowerShell SDK); each has a header explaining **Purpose · How to run · Expected results**. Discovery, Validation, and Report are **read-only**; Remediation and Set-ScopeTargeting **default to** `-WhatIf` and only change anything when you pass `-Apply`.

1. `MACBYOD-Discovery.ps1` — BYOD-readiness discovery: is macOS enrollment allowed, is account-driven

user enrollment present, does a macOS BYOD compliance policy exist, and does a Conditional Access policy gate macOS. Read-only.

1. `MACBYOD-Validation.ps1` — per-device validation for one BYOD Mac: enrolled via **account-driven

user enrollment**,** personally owned**,** compliant**, and Entra-registered — with a clear PASS/FAIL verdict. Read-only.

1. `MACBYOD-Report.ps1` — a BYOD access-posture report: macOS devices by enrollment type (user

enrollment vs device vs unmanaged), compliance state, and which Macs the require-compliance CA policy would gate. Read-only; optional CSV.

1. `MACBYOD-Remediation.ps1` — safe remediation: create/repair the require-compliant-Mac Conditional

Access policy (always re-asserting break-glass exclusions) and assign the BYOD compliance policy. **Defaults to** `-WhatIf`; changes require `-Apply`.

1. `MACBYOD-Set-ScopeTargeting.ps1` — target-group builder/assigner: move the compliance and CA

policies from the pilot group to a production group. **Defaults to** `-WhatIf`; changes require `-Apply`.

Named objects this solution creates

- Pilot security group `BYOD-Mac-Pilot` (Entra)
- Enrollment type **account-driven user enrollment** for macOS (Intune)
- Compliance policy `macOS BYOD Compliance - Pilot` (Intune)
- Conditional Access policy `CA - Require Compliant Mac - Pilot` (Entra)

Hero image

- `MACBYOD-hero.png` — the product image (1200×1200) used on the marketplace listing.

How it's delivered

A **Word** document and **PDF** (cover, table of contents, page numbers, and a scripts appendix), plus all five PowerShell scripts in a single **ZIP** — yours to keep and reuse across **unlimited devices**.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

MACBYOD-Discovery.ps1

Purpose: Confirms the Intune/Entra tenant is ready to onboard personal Macs with the lightest footprint and gate them with Conditional Access. Checks: macOS enrollment allowed, a macOS BYOD compliance policy present, and whether a Conditional Access policy requires a compliant/enrolled Mac. Reads only — makes no changes. Use 'your-tenant-id' style values, never real secrets in scripts. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACBYOD-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK. Graph scopes (read-only): DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, Policy.Read.All. EXPECTED RESULTS A PASS/FAIL line per check and an overall verdict. A READY tenant shows macOS enrollment allowed, a macOS BYOD compliance policy present, and a require-compliant-Mac Conditional Access policy found (report-only or enforced). .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param()

How to run: powershell -ExecutionPolicy Bypass -File .\MACBYOD-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK. Graph scopes (read-only): DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, Policy.Read.All. EXPECTED RESULTS A PASS/FAIL line per check and an overall verdict. A READY tenant shows macOS enrollment allowed, a macOS BYOD compliance policy present, and a require-compliant-Mac Conditional Access policy found (report-only or enforced). .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param()

MACBYOD-Remediation.ps1

Purpose: Performs the common fixes that make the BYOD gate work: creates or repairs the Conditional Access policy that requires a compliant Mac (scoped to macOS, targeting the pilot group) and ALWAYS re-asserts break-glass exclusions so scope changes never drop the lockout safeguard; and assigns the macOS BYOD compliance policy to a target group. Change-making — so it DEFAULTS TO -WhatIf and only writes when you pass -Apply. Use 'your-tenant.onmicrosoft.com' style values, never secrets. HOW TO RUN Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\MACBYOD-Remediation.ps1 -PolicyName "CA - Require Compliant Mac - Pilot" -IncludeGroup BYOD-Mac-Pilot -BreakGlassUpns bg1@your-tenant.onmicrosoft.com,bg2@your-tenant.onmicrosoft.com Apply (creates/repairs the report-only CA policy and re-asserts break-glass): powershell -ExecutionPolicy Bypass -File .\MACBYOD-Remediation.ps1 -PolicyName "CA - Require Compliant Mac - Pilot" -IncludeGroup BYOD-Mac-Pilot -BreakGlassUpns bg1@your-tenant.onmicrosoft.com,bg2@your-tenant.onmicrosoft.com -Apply Requires the Microsoft Graph PowerShell SDK. Graph scopes: read — Policy.Read.All, Group.Read.All, User.Read.All write — Policy.ReadWrite.ConditionalAccess (added only with -Apply). EXPECTED RESULTS Without -Apply: prints the intended CA policy (include group, macOS platform, require-compliant grant, break-glass exclusions) and makes no change. With -Apply: the policy is created/updated in REPORT-ONLY with break-glass excluded — you enable it manually after reviewing sign-in logs. .NOTES White-label — the repeatable way. Defaults to -WhatIf; changes require -Apply. Break-glass always excluded. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$PolicyName, [string]$IncludeGroup = 'BYOD-Mac-Pilot', [string[]]$BreakGlassUpns, [switch]$Apply, [switch]$WhatIf = $true) if ($Apply) {

$WhatIf = $false }

How to run: Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\MACBYOD-Remediation.ps1 -PolicyName "CA - Require Compliant Mac - Pilot" -IncludeGroup BYOD-Mac-Pilot -BreakGlassUpns bg1@your-tenant.onmicrosoft.com,bg2@your-tenant.onmicrosoft.com Apply (creates/repairs the report-only CA policy and re-asserts break-glass): powershell -ExecutionPolicy Bypass -File .\MACBYOD-Remediation.ps1 -PolicyName "CA - Require Compliant Mac - Pilot" -IncludeGroup BYOD-Mac-Pilot -BreakGlassUpns bg1@your-tenant.onmicrosoft.com,bg2@your-tenant.onmicrosoft.com -Apply Requires the Microsoft Graph PowerShell SDK. Graph scopes: read — Policy.Read.All, Group.Read.All, User.Read.All write — Policy.ReadWrite.ConditionalAccess (added only with -Apply). EXPECTED RESULTS Without -Apply: prints the intended CA policy (include group, macOS platform, require-compliant grant, break-glass exclusions) and makes no change. With -Apply: the policy is created/updated in REPORT-ONLY with break-glass excluded — you enable it manually after reviewing sign-in logs. .NOTES White-label — the repeatable way. Defaults to -WhatIf; changes require -Apply. Break-glass always excluded. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$PolicyName, [string]$IncludeGroup = 'BYOD-Mac-Pilot', [string[]]$BreakGlassUpns, [switch]$Apply, [switch]$WhatIf = $true) if ($Apply) { $WhatIf = $false }

MACBYOD-Report.ps1

Purpose: Exports every macOS device in Intune with the fields that matter for BYOD: enrollment model (account-driven user enrollment vs device enrollment vs corporate/supervised), ownership, supervision, compliance state, and last check-in. Flags Macs that would be BLOCKED by a require-compliant Conditional Access gate (non-compliant or unmanaged), so you can see the access posture at a glance. Reads only — makes no changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACBYOD-Report.ps1 powershell -ExecutionPolicy Bypass -File .\MACBYOD-Report.ps1 -CsvPath C:\Temp\macos-byod.csv Requires the Microsoft Graph PowerShell SDK. Graph scope (read-only): DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS A summary (total Macs, user-enrolled BYOD, personally owned, compliant, and would-be-gated counts) plus a CSV with one row per device. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$CsvPath = "$PSScriptRoot\macos-byod-report.csv")

How to run: powershell -ExecutionPolicy Bypass -File .\MACBYOD-Report.ps1 powershell -ExecutionPolicy Bypass -File .\MACBYOD-Report.ps1 -CsvPath C:\Temp\macos-byod.csv Requires the Microsoft Graph PowerShell SDK. Graph scope (read-only): DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS A summary (total Macs, user-enrolled BYOD, personally owned, compliant, and would-be-gated counts) plus a CSV with one row per device. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$CsvPath = "$PSScriptRoot\macos-byod-report.csv")

MACBYOD-Set-ScopeTargeting.ps1

Purpose: Expands the macOS BYOD gate from the pilot group to a production group by (a) assigning the macOS BYOD compliance policy to the target group and (b) setting the require-compliant-Mac Conditional Access policy's included group to the target group while KEEPING break-glass exclusions. Change-making — so it DEFAULTS TO -WhatIf and only writes when you pass -Apply. Break-glass accounts are always re-asserted so scope expansion never drops the lockout safeguard. Use placeholder values like 'your-tenant.onmicrosoft.com', never real secrets. HOW TO RUN Preview (default, no changes):

powershell -ExecutionPolicy Bypass -File .\MACBYOD-Set-ScopeTargeting.ps1 -TargetGroup BYOD-Mac-Users -CompliancePolicyName "macOS BYOD Compliance - Pilot" -CaPolicyName "CA - Require Compliant Mac - Pilot" -BreakGlassUpns bg1@your-tenant.onmicrosoft.com,bg2@your-tenant.onmicrosoft.com Apply the scope change: powershell -ExecutionPolicy Bypass -File .\MACBYOD-Set-ScopeTargeting.ps1 -TargetGroup BYOD-Mac-Users -CompliancePolicyName "macOS BYOD Compliance - Pilot" -CaPolicyName "CA - Require Compliant Mac - Pilot" -BreakGlassUpns bg1@your-tenant.onmicrosoft.com,bg2@your-tenant.onmicrosoft.com -Apply Requires the Microsoft Graph PowerShell SDK. Graph scopes: read — DeviceManagementConfiguration.Read.All, Policy.Read.All, Group.Read.All, User.Read.All write — DeviceManagementConfiguration.ReadWrite.All, Policy.ReadWrite.ConditionalAccess (with -Apply). EXPECTED RESULTS Without -Apply: prints the intended assignments (compliance policy -> target group; CA include -> target group; break-glass still excluded) and makes no change. With -Apply: both assignments are updated and confirmed. .NOTES White-label — the repeatable way. Defaults to -WhatIf; changes require -Apply. Break-glass always excluded. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$TargetGroup, [string]$CompliancePolicyName = 'macOS BYOD Compliance - Pilot', [string]$CaPolicyName = 'CA - Require Compliant Mac - Pilot', [string[]]$BreakGlassUpns, [switch]$Apply, [switch]$WhatIf = $true) if ($Apply) { $WhatIf = $false }

How to run: Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\MACBYOD-Set-ScopeTargeting.ps1 -TargetGroup BYOD-Mac-Users -CompliancePolicyName "macOS BYOD Compliance - Pilot" -CaPolicyName "CA - Require Compliant Mac - Pilot" -BreakGlassUpns bg1@your-tenant.onmicrosoft.com,bg2@your-tenant.onmicrosoft.com Apply the scope change: powershell -ExecutionPolicy Bypass -File .\MACBYOD-Set-ScopeTargeting.ps1 -TargetGroup BYOD-Mac-Users -CompliancePolicyName "macOS BYOD Compliance - Pilot" -CaPolicyName "CA - Require Compliant Mac - Pilot" -BreakGlassUpns bg1@your-tenant.onmicrosoft.com,bg2@your-tenant.onmicrosoft.com -Apply Requires the Microsoft Graph PowerShell SDK. Graph scopes: read — DeviceManagementConfiguration.Read.All, Policy.Read.All, Group.Read.All, User.Read.All write — DeviceManagementConfiguration.ReadWrite.All, Policy.ReadWrite.ConditionalAccess (with -Apply). EXPECTED RESULTS Without -Apply: prints the intended assignments (compliance policy -> target group; CA include -> target group; break-glass still excluded) and makes no change. With -Apply: both assignments are updated and confirmed. .NOTES White-label — the repeatable way. Defaults to -WhatIf; changes require -Apply. Break-glass always excluded. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$TargetGroup, [string]$CompliancePolicyName = 'macOS BYOD Compliance - Pilot', [string]$CaPolicyName = 'CA - Require Compliant Mac - Pilot', [string[]]$BreakGlassUpns, [switch]$Apply, [switch]$WhatIf = $true) if ($Apply) { $WhatIf = $false }

MACBYOD-Validation.ps1

Purpose: For a single macOS device (by serial number or device name), confirms it is: enrolled in Intune via account-driven USER enrollment, personally owned (not corporate/supervised), reporting a compliance state, and registered with Microsoft Entra ID. Produces a clear PASS/FAIL verdict. Reads only — makes no changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACBYOD-Validation.ps1 -Serial C02XXXXXXXXX powershell -ExecutionPolicy Bypass -File .\MACBYOD-Validation.ps1 -DeviceName "Jane-MacBook" Requires the Microsoft Graph PowerShell SDK. Graph scopes (read-only): DeviceManagementManagedDevices.Read.All, DeviceManagementConfiguration.Read.All. EXPECTED RESULTS A per-check PASS/FAIL table and an overall verdict. A correctly onboarded BYOD Mac passes

every check; a whole-device-managed or supervised Mac indicates the wrong enrollment model was used. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$Serial, [string]$DeviceName)

How to run: powershell -ExecutionPolicy Bypass -File .\MACBYOD-Validation.ps1 -Serial C02XXXXXXXX powershell -ExecutionPolicy Bypass -File .\MACBYOD-Validation.ps1 -DeviceName "Jane-MacBook" Requires the Microsoft Graph PowerShell SDK. Graph scopes (read-only): DeviceManagementManagedDevices.Read.All, DeviceManagementConfiguration.Read.All. EXPECTED RESULTS A per-check PASS/FAIL table and an overall verdict. A correctly onboarded BYOD Mac passes every check; a whole-device-managed or supervised Mac indicates the wrong enrollment model was used. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$Serial, [string]$DeviceName)